

Mohammad Mahmudul Hasan

Team Lead, AI Engineer | Computer vision, GIS and photogrammetry

Bangladesh · github.com/fahome10bd · linkedin.com/in/fahome10bd

RESEARCH PROFILE

AI engineering team lead with experience in algorithm development and system design for computer vision and GIS applications. My work spans photogrammetric orthophoto production, point-cloud processing, urban building-change analysis and road-asset inspection. Seeking graduate research in reliable geospatial AI and 3D mapping.

EDUCATION

Islamic University of Technology, Bangladesh — BSc in Electrical and Electronic Engineering, completed 22 March 2021. CGPA **3.76/4.00**; first class with honours.

Selected coursework: Project and Thesis (A+ in both courses); Digital Signal Processing (A); Artificial Neural Networks and Fuzzy Logic (A+).

PROFESSIONAL EXPERIENCE

HawarIT Limited — with the company since 2021; current role **Team Lead, AI Engineer**. Lead a six-member AI engineering team.

- Develop algorithms and design systems for computer vision and GIS tasks using imagery and spatial data.
- Work across AT-related processing, DSM/DTM, image rectification, seamlines, tiling, bridge editing and final orthophoto output.
- Process and align point clouds across multiple trajectories, followed by segmentation and classification.
- Support GIS digitization, multi-year building-change analysis and traffic-sign assessment from car-captured 360-degree imagery.

PEER-REVIEWED PUBLICATION

Tasnim Sakib Apon, **Mohammad Mahmudul Hasan**, Abrar Islam, and Md. Golam Rabiul Alam. "Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI." 2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE). doi:[10.1109/CSDE53843.2021.9718400](https://doi.org/10.1109/CSDE53843.2021.9718400).

SELECTED RESEARCH AND PROJECTS

EEG undergraduate thesis: signal analysis related to impaired consciousness and coma.

Building change and mapping: interpretation of construction, demolition and extensions from multi-year orthophotos; GIS digitization from imagery.

Traffic detection: earlier 21-label computer-vision project with a [public repository](#).

TECHNICAL AREAS AND LANGUAGE

Computer vision; GIS; algorithm development; system design; photogrammetric workflows; point-cloud alignment and classification; signal analysis; Python; MATLAB.

Bangla: native. English-medium instruction certificate available. IELTS pending.